WORKED SOLUTION

STEP III 1998, Question 12

Probability and Statistics • Network Reliability

STEP III 1998, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q12

The key idea

There are only six roads, so there are only $2^6 = 64$ patterns of open and blocked roads, and the cleanest route is to count the *bad* patterns: those in which the villages split into groups that can't reach each other. Sorting the bad patterns by how many roads are still open makes the count very short, because a network with four or more open roads on four villages can never be disconnected.

A warning before we start: the formula as printed in the paper has p^2 in front of the bracket, but the correct factor is p^3. The printed version can't be right, as we show in the Checks. We prove the corrected result

$$P = 1 - p^3(6p^3 - 12p^2 + 3p + 4).$$

Solution

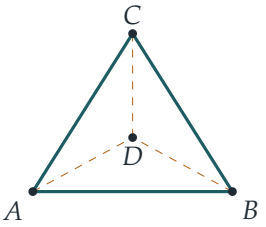
Write $q = 1 - p$ for the probability that a given road is open. Since the roads behave independently, any particular pattern with k roads open and $6 - k$ blocked has probability $q^k p^{6-k}$. Let D be the event that the network is *disconnected*, meaning some village can't reach some other village. Then $P = 1 - \mathbb{P}(D)$, and $\mathbb{P}(D)$ is the sum of $q^k p^{6-k}$ over all disconnected patterns. We count these according to k, the number of open roads.

Four or more open roads: never disconnected. If the network is disconnected, the villages split into at least two groups with no open road between groups. The only possible splits are $1 + 3$ (a lone village plus the other three) and $2 + 2$. All open roads lie inside the groups. In a $1 + 3$ split the group of three has only 3 roads; in a $2 + 2$ split each pair has only 1 road, so 2 in total. Either way at most 3 roads can be open. So every pattern with $k \geq 4$ open roads is connected.

Exactly three open roads.

Ask yourself With three open roads, which patterns fail to connect all four villages?

By the argument above, a disconnected pattern with three open roads must be a $1+3$ split in which all three roads among the group of three are open: a *triangle*, with the fourth village cut off. There are 4 choices of the isolated village, so 4 such patterns. (The other $\binom{6}{3} - 4 = 16$ patterns are the spanning trees of the tetrahedron, and they are connected.)



open roads AB, BC, CA (solid); blocked roads to D (dashed): D is cut off

Two or fewer open roads: always disconnected. To link four villages we need at least three roads (each road can join at most one new village to the group already reachable from A). So all $\binom{6}{2} = 15$ patterns with two open roads, all 6 with one, and the 1 with none are disconnected.

Adding up. Therefore

$$\mathbb{P}(D) = 4p^3q^3 + 15p^4q^2 + 6p^5q + p^6 = p^3\big(4q^3 + 15pq^2 + 6p^2q + p^3\big).$$

Now expand the bracket with $q = 1 - p$:

$$\begin{aligned} 4(1-p)^3 &= 4 - 12p + 12p^2 - 4p^3, \\ 15p(1-p)^2 &= 15p - 30p^2 + 15p^3, \\ 6p^2(1-p) &= 6p^2 - 6p^3, \\ p^3 &= p^3. \end{aligned}$$

Adding, the constant term is 4, the p terms give $-12p + 15p = 3p$, the p^2 terms give $12 - 30 + 6 = -12$, and the p^3 terms give $-4 + 15 - 6 + 1 = 6$. So

$$\mathbb{P}(D) = p^3\big(6p^3 - 12p^2 + 3p + 4\big), \qquad \text{hence} \qquad P = 1 - p^3\big(6p^3 - 12p^2 + 3p + 4\big).$$

Checks

- $p = 0$ (no road is ever blocked) gives $P = 1$, and $p = 1$ gives $P = 1 - (6 - 12 + 3 + 4) = 0$. Both are right, but both hold for the p^2 version too, so they can't tell the two apart.

- Small p separates them. Disconnection needs at least three blocked roads (the three at one village, say), so $\mathbb{P}(D)$ should be about $4p^3$ when p is small. Our formula gives $\mathbb{P}(D) \approx 4p^3$; the printed one would give $4p^2$, which is impossible.
- At $p = \frac{1}{3}$ our formula gives $P = 1 - \frac{1}{27} \cdot \frac{105}{27} = \frac{208}{243} \approx 0.856$. A computer enumeration of all 64 patterns at 21 values of p agrees exactly with the p^3 formula, and a simulation of 200 000 snowfalls at $p = 0.4$ gave 0.767 against the formula's 0.7655.

Common slips

- Counting only the cases with an isolated village and forgetting to check the $2+2$ splits. Here they add nothing new: a $2+2$ split leaves at most two open roads, so those patterns are already in the "two or fewer" count. You still need to say so.
- Adding up probabilities of overlapping events such as "A is cut off" and "B is cut off" without correcting for the overlap. Counting whole patterns avoids double counting.
- Trusting the printed p^2 and forcing the algebra to fit. If honest working disagrees with a "show that", test a special case.

Beyond STEP

This is a small case of *network reliability*: each edge of a graph fails independently, and we want the probability that the surviving graph is connected. The answer is always a polynomial in p whose coefficients count connected spanning subgraphs (here $1, 6, 15, 16$ for $6, 5, 4, 3$ open roads). For large graphs it is computationally hard to find, and closely tied to the Tutte polynomial and to percolation theory. Grimmett and Welsh, *Probability: An Introduction*, is a good place to build the probability side; Bollobás, *Modern Graph Theory*, covers random graphs and the Tutte polynomial.

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